

Simulation-Based Inference

Approximate Bayesian Computation

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Rejection Sampling (brute force independent sampling)

Choose a total number of T draws and run the algorithm below for all $t = 1, \dots, T$

- Sample parameter values from the prior: $\theta^{(t)} \sim p(\theta)$
- Sample data based on $\theta^{(t)}$ from the likelihood: $y^{(t)} \sim p(y \mid \theta^{(t)})$
- only keep $\theta^{(t)}$ if its data $y^{(t)}$ is equal to the observed data $\tilde{y}$
- Otherwise throw $\theta^{(t)}$ away (reject it)

The S kept parameter values $\theta^{(s)}$ are draws from the posterior: $\theta^{(s)} \sim p(\theta \mid y)$

This is a purely simulation-based inference procedure as no density evaluations are required!

Rejection sampling: Beta-binomial example

Observed response $\tilde{y} = 4$ out of $N = 10$ individuals

Hypothetical iteration $t = 1$:

- Sampling from $\text{beta}(1, 1)$ prior yields $\theta^{(1)} = 0.7$.
- Sampling from the $\text{binomial}(N = 10, \theta = \theta^{(1)})$ likelihood yields $y^{(1)} = 6$.
- The sample response $y^{(1)} = 6$ is different from the observed response $\tilde{y} = 4$ so that $\theta^{(1)} = 0.7$ is thrown away.

Hypothetical iteration $t = 2$:

- Sampling from the $\text{beta}(1, 1)$ prior yields $\theta^{(2)} = 0.42$.
- Sampling from the $\text{binomial}(N = 10, \theta = \theta^{(2)})$ likelihood yields $y^{(2)} = 4$.
- The sample response $y^{(2)} = 4$ is equal to the observed response $\tilde{y} = 4$ so that $\theta^{(2)} = 0.42$ is kept.

Rejection sampling: Beta-binomial example

Specify the observed data:

```
N <- 10  
y_obs <- 4
```

Number of random draws from the prior:

```
ndraws <- 10000
```

Sample `ndraws` draws from the prior:

```
prior <- rbeta(ndraws, 1, 1)
```

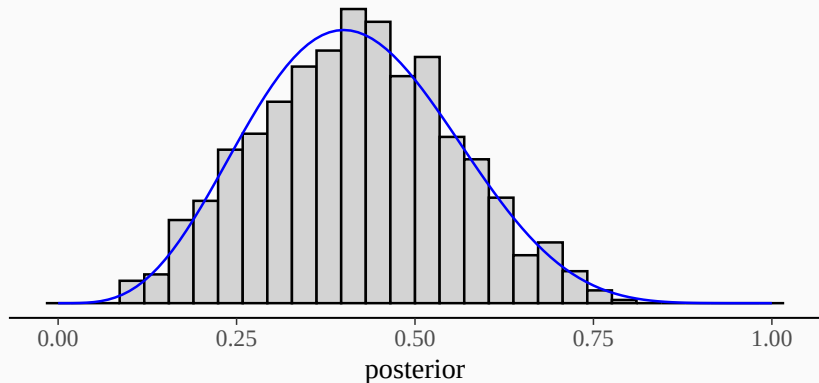
Sample data from the likelihood:

```
y_sim <- rbinom(ndraws, size = N, prob = prior)
```

Filter out all draws that do not match the data:

```
posterior <- prior[y_sim == y_obs]
```

We retain 898 (independent) posterior draws.



Black histogram: Rejection-sampling posterior draws. Blue line: Analytic posterior density: $\text{beta}(5, 7)$.

Rejection sampling implements a distribution $p_{\text{RS}}(\theta \mid \tilde{y})$ for which holds

$$\begin{aligned} p_{\text{RS}}(\theta \mid \tilde{y}) &\propto \int p(\theta) p(y \mid \theta) I(y = \tilde{y}) dy \\ &= p(\theta) \int p(y \mid \theta) I(y = \tilde{y}) dy \\ &= p(\theta) p(\tilde{y} \mid \theta) \\ &\propto p(\theta \mid \tilde{y}) \end{aligned}$$

Accordingly $p_{\text{RS}}(\theta \mid \tilde{y}) = p(\theta \mid \tilde{y})$, the (true) posterior given data $\tilde{y}$.

Rejection sampling for bigger data

Suppose we have surveyed $N = 100$ people of which $\tilde{y} = 40$ have the genetic variant of interest.

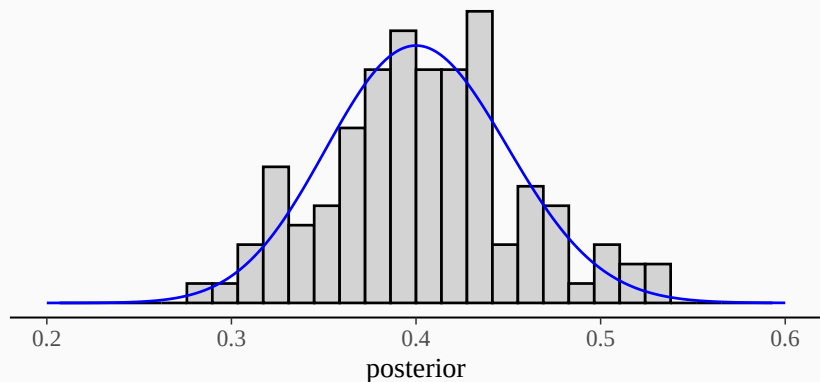
Perform rejection sampling (with the same prior samples):

```
N <- 100
y_obs <- 40
y_sim <- rbinom(length(prior), size = N, prob = prior)
posterior <- prior[y_sim == y_obs]
```

We retain 117 posterior draws.

Sampled vs. analytic posterior

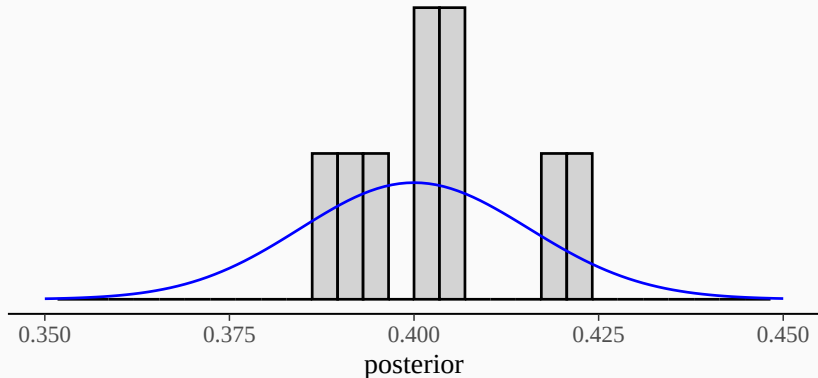
Resulting posterior:



Black histogram: Rejection-sampling posterior draws. Blue line: Analytic posterior density: $\text{beta}(41, 61)$.

Rejection sampling for even bigger data

Suppose we have surveyed $N = 1000$ people of which $\tilde{y} = 400$ have the genetic variant of interest. We retain 9 posterior draws.



Black histogram: Rejection-sampling posterior draws. Blue line: Analytic posterior density: $\text{beta}(401, 601)$.

Approximate Bayesian Computation (ABC)

Basic ABC is essentially rejection sampling where the equality of observed and simulated data only needs to hold *approximately* to accept a sample.

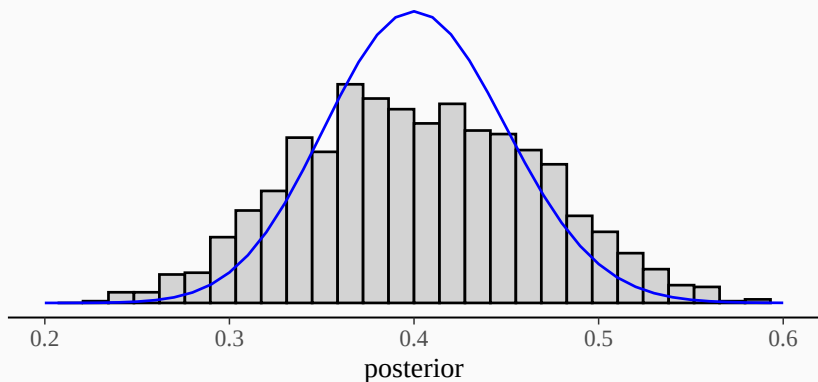
- Choose a distance measure d and a small $\epsilon > 0$.
- Accept all $\theta^{(t)}$ for which $d(y^{(t)}, \tilde{y}) \leq \epsilon$.

ABC for the beta-binomial example: Suppose we set $\epsilon = 0.07N$ as 7% of the sample size N :

- $N = 10$: Exact rejection sampling since $\lfloor \epsilon \rfloor = \lfloor 0.7 \rfloor = 0$.
- $N = 100$: $\epsilon = 7$ so accept all $y^{(t)} \in [y - 7, y + 7]$.
- $N = 1000$: $\epsilon = 70$ so accept all $y^{(t)} \in [y - 70, y + 70]$.

ABC for the beta-binomial example: $N = 100$

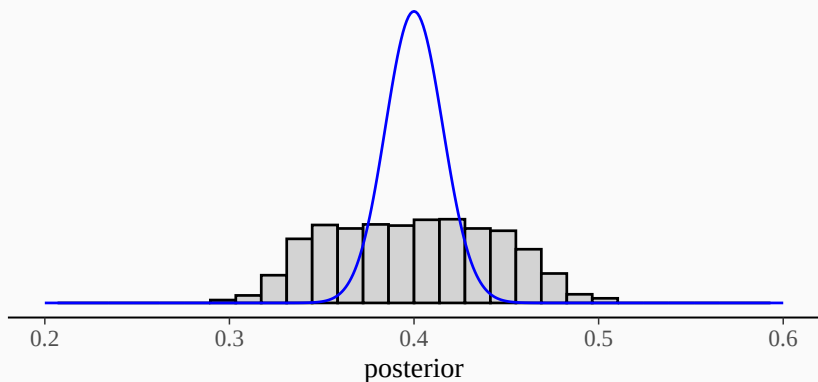
We retain 1451 posterior draws.



Black histogram: ABC posterior draws. Blue line: Analytic posterior density: $\text{beta}(41, 61)$.

ABC for the beta-binomial example: $N = 1000$

We retain 1421 posterior draws.



Black histogram: ABC posterior draws. Blue line: Analytic posterior density: $\text{beta}(401, 601)$.

ABC for continuous data models

For a model with continuous (real) data y , the event $y^{(t)} = \tilde{y}$ has probability 0, so there is no alternative to $\epsilon > 0$ in the first place.

Example: Estimate the mean from normally distributed data.

Prior:

$$\mu \sim \text{normal}(0, 1)$$

Likelihood for each $i = 1, \dots, N$:

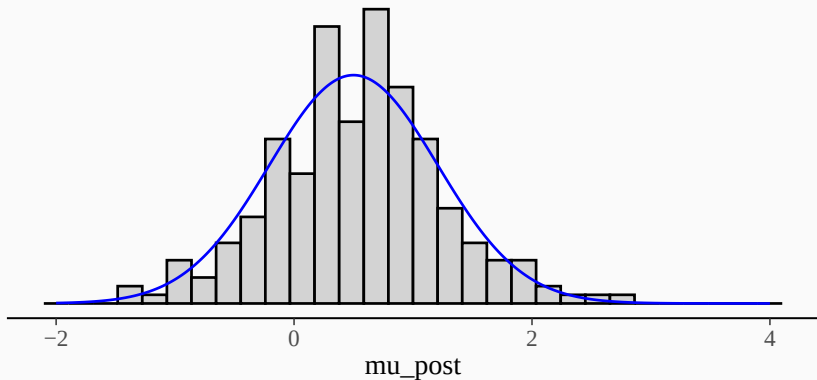
$$y_i \sim \text{normal}(\mu, 1)$$

ABC algorithm for the normal means problem:

- Sample T draws $\mu^{(t)}$ from the prior $\text{normal}(0, 1)$
- Sample T sets of draws $\{y_i^{(t)}\}_{i=1}^N$ from the likelihood $\text{normal}(\mu^{(t)}, 1)$
- Accept $\mu^{(t)}$ if $d(y_i^{(t)}, \tilde{y}_i) \leq \epsilon$ for all $i = 1, \dots, N$

Normal means problem: $N = 1$ observation

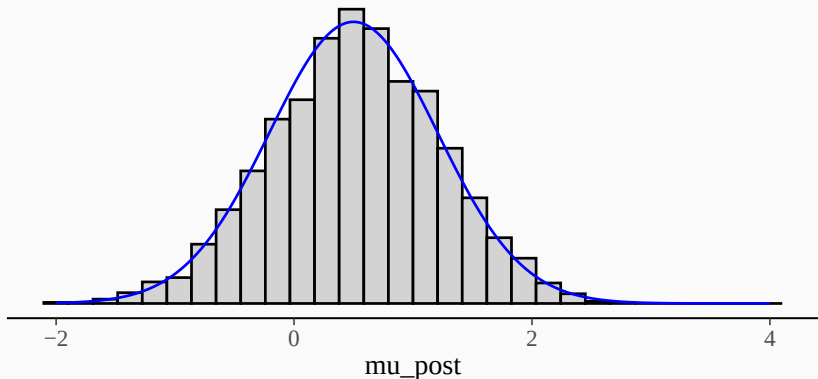
Scenario: $N = 1$ with observed data $\tilde{y}_1 = 1$. We use $S = 10,000$ and $\epsilon = 0.05$.



Black histogram: 226 ABC posterior draws. Blue line: Analytic posterior density.

Normal means problem: $N = 1$ observation

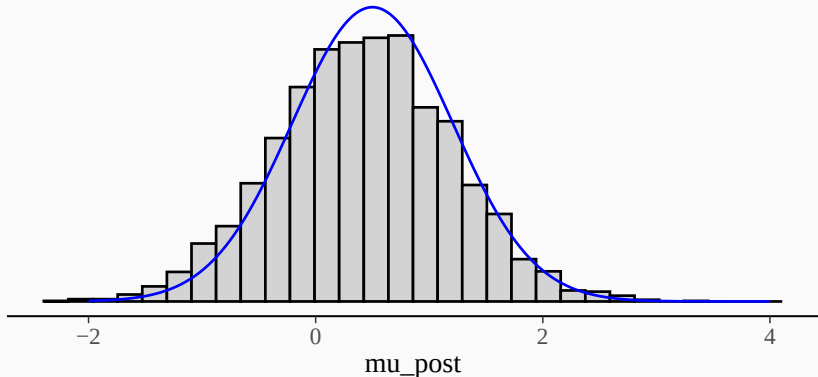
Same data but we use $S = 100,000$ and $\epsilon = 0.05$ for ABC.



Black histogram: 2238 ABC posterior draws. Blue line: Analytic posterior density.

Normal means problem: $N = 1$ observation

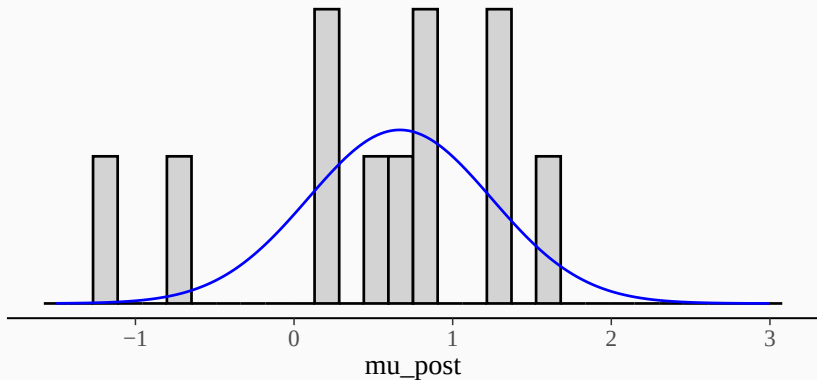
Same data but we use $S = 10,000$ and $\epsilon = 1$ for ABC.



Black histogram: 4153 ABC posterior draws. Blue line: Analytic posterior density.

Normal means problem: $N = 2$ observations

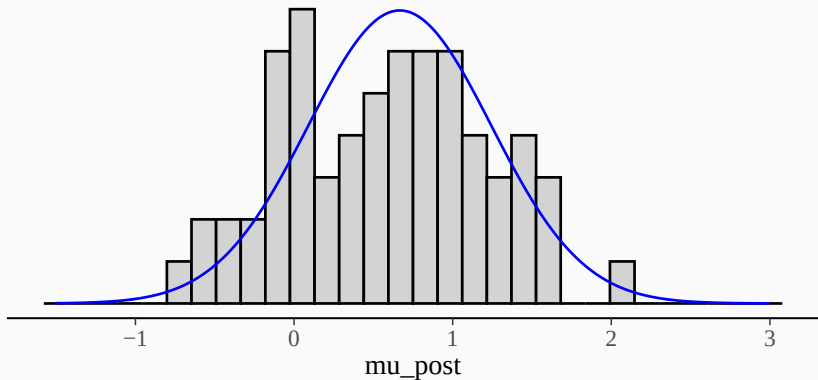
Scenario: $N = 2$ with observed data $\tilde{y} = (1, 1)$. We use $S = 10,000$ and $\epsilon = 0.05$.



Black histogram: 11 ABC posterior draws. Blue line: Analytic posterior density.

Normal means problem: $N = 2$ observations

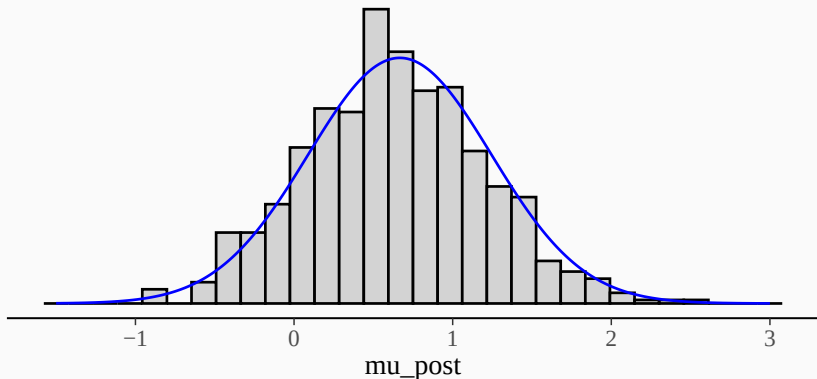
Same data but we use $S = 100,000$ and $\epsilon = 0.05$.



Black histogram: 65 ABC posterior draws. Blue line: Analytic posterior density.

Normal means problem: $N = 2$ observations

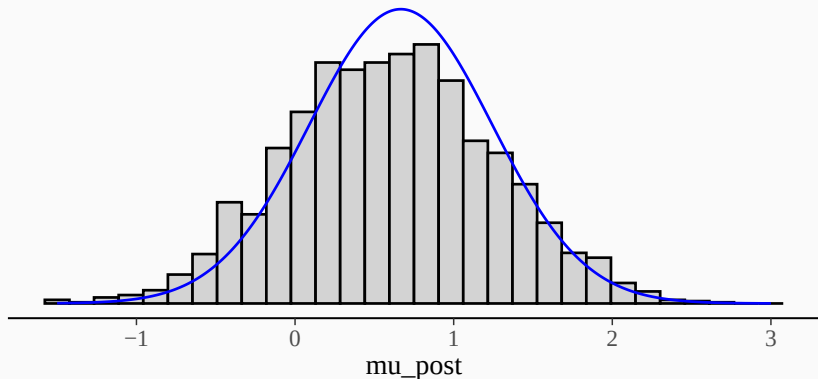
Same data but we use $S = 1,000,000$ and $\epsilon = 0.05$.



Black histogram: 646 ABC posterior draws. Blue line: Analytic posterior density.

Normal means problem: $N = 2$ observations

Same data but we use $S = 10,000$ and $\epsilon = 1$.



Black histogram: 2278 ABC posterior draws. Blue line: Analytic posterior density.

Rejecting based on the individual y_i , if $d(y_i^{(t)}, \tilde{y}_i) > \epsilon$ for **any** $i = 1, \dots, N$, quickly leads to very high rejection rates as N increases.

Solution: Perform the rejection step based on only a small, fixed number of summary statistics $T_j(y_1, \dots, y_N)$ for $j = 1, \dots, J$. We write $T(y)$ for the whole set of summary statistics

Implication: We no longer estimate the posterior $p(\theta \mid y)$ given the original data but the posterior $p(\theta \mid T(y))$ given the summary statistics.

Question: When is $p(\theta \mid T(y))$ close to $p(\theta \mid y)$?

A vector of statistics $T(y)$ is *sufficient* for θ if all the information about θ contained in y is also contained in $T(y)$.

More formally: $T(y)$ is sufficient for θ if $p(y \mid T(y), \theta) = p(y \mid T(y))$.

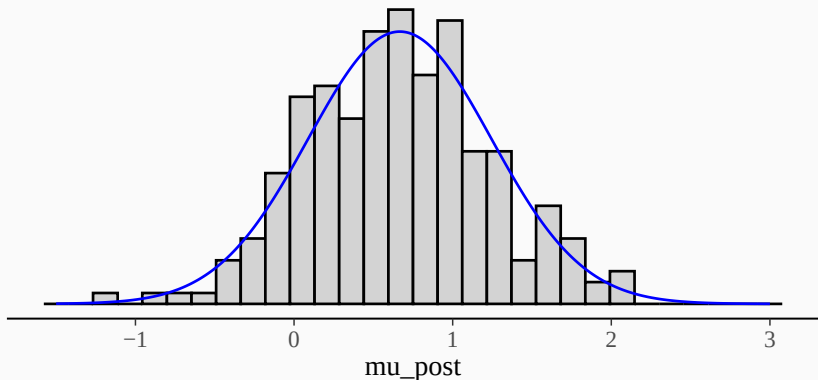
Representation theorem: If $T(y)$ is sufficient, we can write $p(y \mid \theta) = h(y) g(T(y), \theta)$ for some functions h and g .

Implications for Bayes: If $T(y)$ is sufficient then

$$\begin{aligned} p(\theta \mid y) &\propto p(y \mid \theta) p(\theta) = h(y) g(T(y), \theta) p(\theta) \\ &\propto g(T(y), \theta) p(\theta) \propto p(T(y) \mid \theta) p(\theta) \\ &\propto p(\theta \mid T(y)) \end{aligned}$$

Normal means problem with summaries: $N = 2$ observations

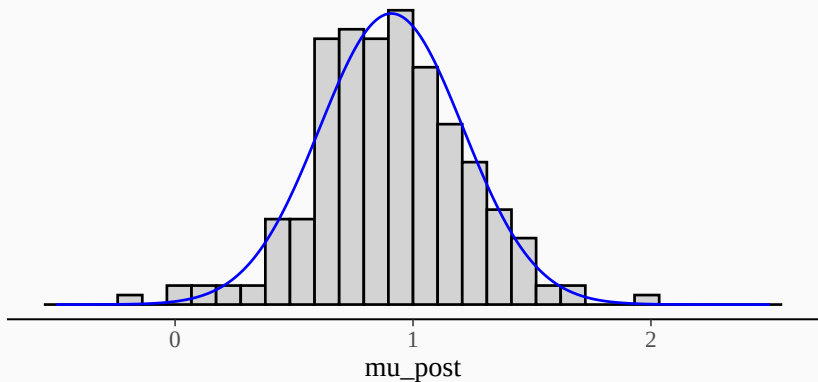
Scenario: $N = 2$ with observed data $\tilde{y} = (1, 1)$. We use the empirical mean $T(y) = \bar{y}$ as a single summary statistic, $S = 10,000$, and $\epsilon = 0.05$.



Black histogram: 233 ABC posterior draws. Blue line: Analytic posterior density. Already $S = 10,000$ produces enough posterior draws.

Normal means problem with summaries: 10 observations

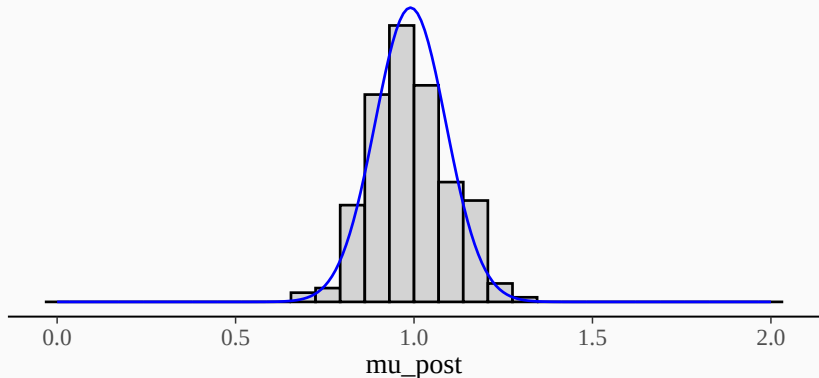
We now check for a larger dataset: $N = 10$ with observed data $\tilde{y} = (1, 1, \dots, 1)$.



Black histogram: 224 ABC posterior draws. Blue line: Analytic posterior density.
Still $S = 10,000$ produces enough posterior draws.

Normal means problem with summaries: 100 observations

We now check for a larger dataset: $N = 100$ with observed data $\tilde{y} = (1, 1, \dots, 1)$.



Black histogram: 231 ABC posterior draws. Blue line: Analytic posterior density. Still $S = 10,000$ produces enough posterior draws.

ABC implements a distribution $p_{\text{ABC}}(\theta \mid \tilde{y})$ for which holds

$$p_{\text{ABC}}(\theta \mid \tilde{y}) \propto \int p(\theta) p(y \mid \theta) I(d(T(y), T(\tilde{y})) \leq \epsilon) dy$$

If $T(y)$ is a sufficient statistic and $\epsilon \rightarrow 0$, this simplifies to

$$\begin{aligned} p_{\text{ABC}}(\theta \mid \tilde{y}) &\propto \int p(\theta) p(T(y) \mid \theta) I(d(T(y), T(\tilde{y})) \leq \epsilon) dy \\ &= p(\theta) \int p(T(y) \mid \theta) I(T(y) = T(\tilde{y})) dy \\ &= p(\theta) p(T(\tilde{y}) \mid \theta) \\ &\propto p(\theta \mid T(\tilde{y})) \\ &= p(\theta \mid \tilde{y}) \end{aligned}$$

(Start) Choose a total number of draws T and an initial value $\theta^{(1)}$.

Run the steps below for all $t = 1, \dots, T$:

(1) Sample $\theta^{(p)}$ from a proposal distribution $q(\theta^{(p)} \mid \theta^{(t)})$.

(2) Sample proposal data $y^{(p)} \sim p(y \mid \theta^{(p)})$.

(3) If $d(T(y^{(p)}), T(\tilde{y})) \leq \epsilon$:

- Compute the ratio

$$\alpha = \min \left(1, \frac{p(\theta^{(p)}) q(\theta^{(t)} \mid \theta^{(p)})}{p(\theta^{(t)}) q(\theta^{(p)} \mid \theta^{(t)})} \right)$$

- Set $\theta^{(t+1)} = \theta^{(p)}$ with probability α and $\theta^{(t+1)} = \theta^{(t)}$ otherwise.

(4) Else, set $\theta^{(t+1)} = \theta^{(t)}$.

(End) Throw away the first t^* draws as warmup.

ABC and its variants are fully simulation-based procedures that do not require knowledge of the likelihood density. Some variants don't require the prior density either.

While very general, even modern variants of ABC struggle to overcome the curse of dimensionality in models with many parameters. This is because, to estimate many parameters, many summary statistics are required, leading to extremely high rejection rates during sampling.

This issue is in some sense “unfixable” within ABC. Hence, we will consider fundamentally different approaches to Bayesian SBI in the upcoming weeks.